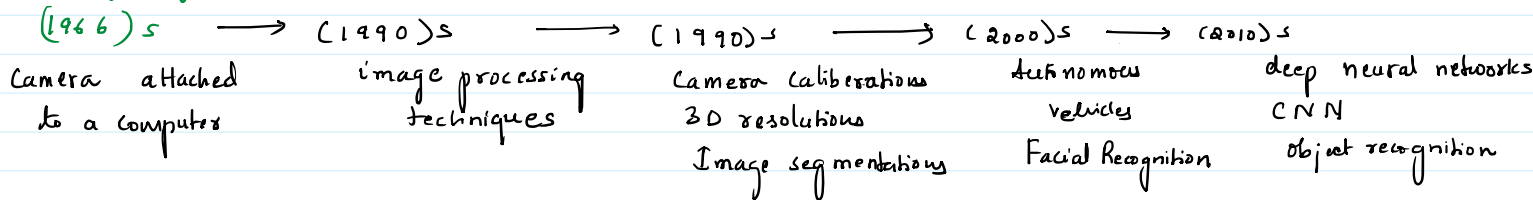


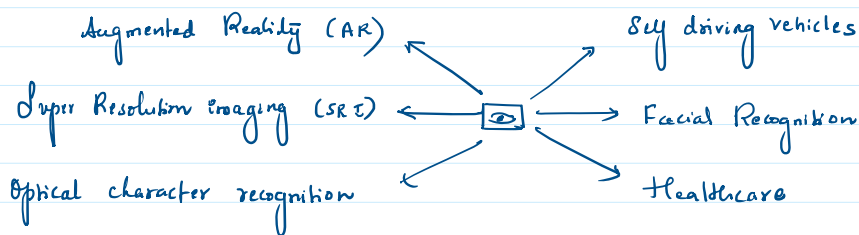
enables - machines to understand, analyze and interpret visual information same way humans do.

eg: with CV it trains machines to perform actions digitally.

History of Computer Vision :-



Computer vision applications:



Computer vision today

- * Continuous implement in neural networks.
- * availability of cameras and sensors.
- * frameworks that enable processing of images - (edge detection)
- * medical image analysis, self driving cars.
- * Security through video surveillance,
- * automation → robots and manufacturing.
- * increases reliability, affordability, performance, functionality of computer vision technologies.

How does it work:

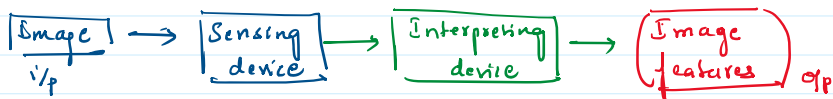


Image Acquisition



Preprocessing



feature extraction



object classification



object identification



Object Tracking

Process of CV

Object Tracking

Application of CV

Healthcare

Financial Services

Government

Retail and e-commerce

Image fundamentals:

- 1) A digital image $\rightarrow$ 2d array of pixels
- 2) Color spaces $\rightarrow$ RGB
- 3) Grayscale
- 4) HSV (Hue, Saturation, Value)
- 5) Resolution: width $\times$ height
- 6) Channels: no of color planes

Image transformations:

- ① Geometric: Rotate, Scale, flip
- ② Filtering: blur, sharpen, edge detection
- ③ Intensity: normalization, thresholding